WEEK 03

Collections, loops and totals

Research 2 exchange · Class warm-up before Session 1

Python foundations

Collections, loops and totals

Lists, records and accumulation

Lists hold ordered items

Index 0 selects the first item; `len` counts items, not their sum.

```
scores = [2, 4]
print(scores[0], len(scores))  # 2 2
```

Dictionaries name fields

A key selects one value in a record.

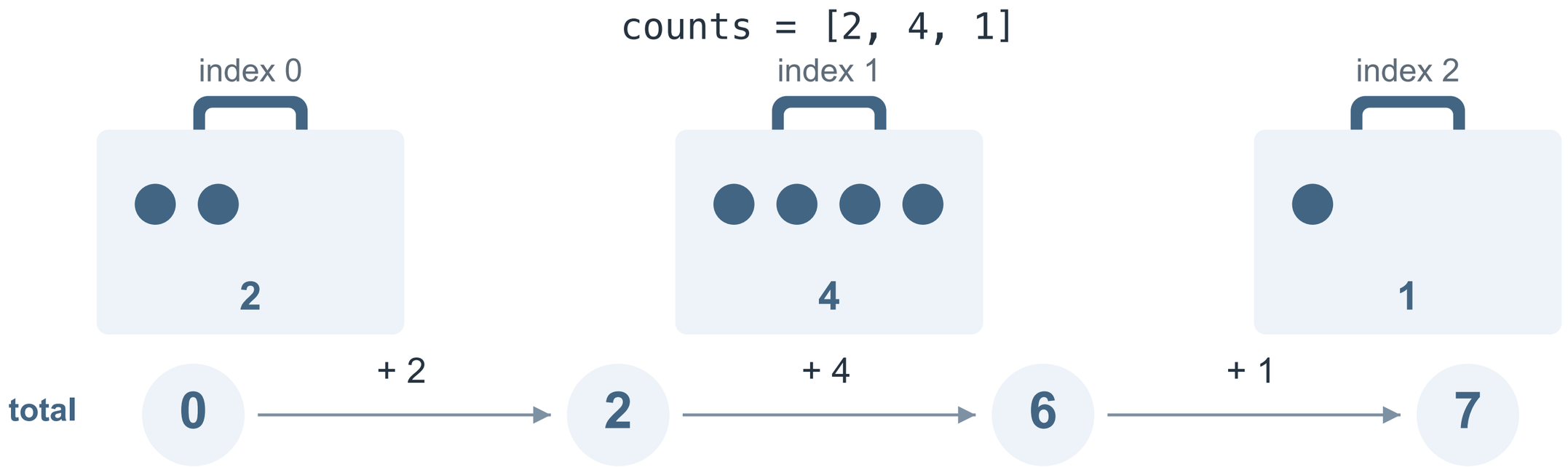
```
row = {"name": "A", "points": 2}
print(row["points"])  # 2
```

Loops accumulate

Initialize once, then add each item. Empty input leaves zero.

```
total = 0
for value in [2, 4]: total += value
```

Three bags, seven pieces of fruit




```
len(counts)  # 3 bags     |     total  # 7 pieces of fruit
```

Initialize once. Each visit adds to the earlier total.

Syntax to read and reuse

Python / syntax	Meaning	When to use it
[item, item]	List: ordered items	Store several records
{"key": value}	Dictionary: named fields	Describe one record
for item in items:	Visit each item	Process a collection
total += value	Add to the running total	Accumulate contributions

Python Basics · sections 05, 06 · use only when you need a lookup.

Create, count and read a plain list

```
counts = [2, 4, 1]
print(len(counts))
print(counts[0])
print(counts[1])
empty = []
print(len(empty))
```

1. A list stores items in order.

Update, append and remove an item

```
01 counts = [2, 4]
02 counts[1] = 3
03 print(counts)
04 counts.append(1)
05 print(counts)
06 last = counts.pop()
07 print(last, counts)
```

1. Assigning counts[1] replaces one item.

Visit a plain list with a loop

```
counts = [2, 4, 1]
for count in counts:
    print(count)
print("Finished")
```

1. In for count in counts, count is one number at a time.

Accumulate numbers before records

```
counts = [2, 4, 1]
total = 0
for count in counts:
    total += count
    print("Running:", total)
print("Final:", total)
assert total == 7
```

1. Initialize total once before the loop, then update it for each number.

Read one dictionary before a list of dictionaries

```
record = {"name": "A", "points": 2}
print(record["name"])
print(record["points"])
record["points"] = 3
print(record["points"])
```

1. A dictionary uses named keys, not positions.

Read fields and accumulate a total

```
records = [{"name": "A", "count": 2},
           {"name": "B", "count": 3}]
print(records[0]["name"])
total = 0
for row in records:
    total += row["count"]
    print(row["name"], total)
print("total", total)
```

1. Expected: A; A 2; B 5; total 5.
2. The accumulator starts before the loop.
3. Try an empty list: what still runs?

Lists contain items; dictionaries contain fields

1. **List** — Ordered items. A position selects one item; indexing starts at zero.
2. **Dictionary** — Named key/value pairs. A key selects one field.
3. **Value** — The selected value can be used in an expression or another statement.

A list position and a dictionary key select different things.

Read a dictionary value by key

```
line = {"batch_id": "T01",
        "portions_per_container": 3, "containers": 4}
print(line["batch_id"])
print(line["containers"])
planned = line["portions_per_container"]
planned *= line["containers"]
print("Planned portions:", planned)
assert planned == 12
```

1. Predict the ID, count and labelled portion estimate
2. Keys retrieve named values
3. A wrong key spelling raises KeyError

The estimate combines two fields from the same record.

for: visit each item in a list

```
lines = [
    {"batch_id": "T01", "containers": 4},
    {"batch_id": "T02", "containers": 2},
]
for line in lines:
    print(line["batch_id"], line["containers"])
print("Lines:", len(lines))
assert len(lines) == 2
```

1. Write the output in execution order.
2. The indented print runs for each item.
3. The final print runs after the loop.

The loop body runs once per item; len() counts items.

Trace the accumulator after each contribution

```
contributions = [12, 8]
total = 0
for contribution in contributions:
    before = total
    total += contribution
    print(before, "+", contribution, "=", total)
assert total == 20
```

1. Trace the first contribution from the initial zero
2. Use that running total for the second contribution
3. The previous total is part of the next calculation

Initialize once, add each contribution, then report.

An empty list: zero loop iterations

```
contributions = []
planned_portions = 0
for contribution in contributions:
    planned_portions += contribution
print("Planned portions:", planned_portions)
assert planned_portions == 0
```

1. How often does the body execute?
2. Which value remains afterward?
3. Initialize before the loop, even for [].

An empty list skips the loop body and keeps the initial value.

Hands-on practice

Week_03.ipynb · Session 2

Python practice

P1–P2 · Core

Two small programs with approach hints.

P3 · Core

A complete program with normal and edge-case tests.

P4–P5 · Optional

An application and a harder challenge.

P1–P3: your code, expected results and an explanation. P4–P5 are optional.

Project studio

Multi-batch aggregation

The project need

1

Need

A pickup can contain more than one kind of food.

2

Operation

Read each food line and add its containers and planned portions.

3

Result

The coordinator can read the totals without mixing lines, containers and portions.

Final demo: begin with two counts; expand to records only when needed.

Test a split without changing the underlying plan

1

Original

One four-container T01 line

2

Transform

Two two-container T01 lines with the same declaration

3

Predict

One extra line; unchanged container and portion totals

4

Check

Compare both representations with independent expectations

Line count may change while containers and planned portions stay equal.

Project task

Supplied starting point

The notebook supplies BREAD-A and RICE-A line dictionaries.

You run / explain

Read each line, multiply containers by unit size, and accumulate container and portion totals outside/inside the loop correctly.

Check the evidence

Starter: 2 lines, 5 containers, 16 portions. Empty list → zeros. Split the bread line: total portions must stay 16.

Explain one field choice and trace one contribution into the total.

Project checks

Step	What to show
Implementation	Loop over lines; accumulate containers and portions.
Checks	Starter 2/5/16; empty → zeros; split line preserves total.

Explain one field choice and trace one contribution into the total.

Further reading in English

1. ## Python tutorial: data structures

   https://docs.python.org/3/tutorial/datastructures.html

2. ## Python tutorial: for statements

   https://docs.python.org/3/tutorial/controlflow.html#for-statements

Read one relevant section, then bring an example and a question.